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THERMALLY-CONDUCTIVE CRYSTALLINE PEDESTAL FOR SEMICONDUCTOR DEVICE PACKAGES

Abstract

A semiconductor device package includes a semiconductor die having two opposing faces that define a major plane, wherein the semiconductor die generates heat when in operation. The package includes a packaging lid, contacts, a crystalline pedestal, and a layer of thermal interface material (TIM). The packaging lid encloses the semiconductor die. The contacts are on a first exterior surface of the package parallel to the major plane, the first exterior surface defining a bottom of the package. The crystalline pedestal is formed of one or more crystals having an anisotropic thermal property affecting a thermal conductivity of the pedestal to dissipate the heat generated by the die when in operation, and the pedestal is disposed above the die in thermally-conductive, electrically non-conductive contact with the semiconductor die. The layer of TIM is disposed between the pedestal and the lid, wherein the TIM is thermally conductive and electrically non-conductive.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION [0001] This disclosure claims the benefit of commonly-assigned U.S. Provisional Patent Application No. 63/563,298, filed Mar. 8, 2024, which is hereby incorporated by reference herein in its entirety.

FIELD OF USE

[0002] This disclosure relates to the packaging of a semiconductor die. More particularly, the disclosure relates to packages and methods, including a semiconductive thermal crystalline pedestal in thermal contact with the semiconductor die.

BACKGROUND

[0003] The background description provided herein is for the purpose of generally presenting the context of the disclosure. Work of the inventors hereof, to the extent the work is described in this background section, as well as aspects of the description that may not otherwise qualify as prior art at the time of filing, are neither expressly nor impliedly admitted to be prior art against the subject matter of the present disclosure.

[0004] A semiconductor device includes a semiconductor die on which the device circuits are formed. Typically, the semiconductor die is protected by packaging (including, e.g., a packaging lid). Within the semiconductor device package, the semiconductor die may be mounted on a supporting substrate with conductive interconnections. Leads connected to terminals of the semiconductor die are coupled to contacts on the outside of the semiconductor device package. The contacts may take the form of rigid metallic pins or, alternatively, conductive bumps or solder balls. The semiconductor die may be completely enclosed in the semiconductor device package, or one surface (e.g., where the semiconductor die is a rectangular solid, one of the two largest surfaces) may be exposed. The semiconductor device package should allow for dissipation of heat generated by the semiconductor die when in operation.

SUMMARY

[0005] In accordance with implementations of the subject matter of this disclosure, a semiconductor device package including a semiconductor die having two opposing faces that define a major plane, wherein the semiconductor die generates heat when in operation, a packaging lid enclosing the semiconductor die, a plurality of contacts on a first exterior surface of the semiconductor device package that is parallel to the major plane, the first exterior surface defining a bottom of the semiconductor device package, a crystalline pedestal formed of one or more crystals having an anisotropic thermal property affecting a thermal conductivity of the crystalline pedestal to dissipate the heat generated by the semiconductor die when in operation, the crystalline pedestal disposed above the semiconductor die in thermally-conductive, electrically non-conductive contact with the semiconductor die, and a first layer of a thermal interface material disposed between the pedestal and the packaging lid, wherein the thermal interface material is thermally conductive and electrically non-conductive.

[0006] In a first implementation of such a semiconductor device package, the semiconductor device package may further include a substrate layer below the semiconductor die and forming the first exterior surface.

[0007] In a second implementation of such a semiconductor device package, the semiconductor device package may further include a second layer of the thermal interface material disposed between (i) the semiconductor die and (ii) the crystalline pedestal, wherein the second layer of the thermal interface material is configured to promote transfer of the heat from the semiconductor die to the crystalline pedestal.

[0008] In a third implementation of such a semiconductor device package, the one or more crystals of the crystalline pedestal may be oriented to dissipate the heat in a direction extending from the semiconductor die toward the packaging lid.

[0009] According to a first aspect of that third implementation, the one or more crystals of the crystalline pedestal may be oriented in a 111 crystallographic direction.

[0010] In a fourth implementation of such a semiconductor device package, heat-producing circuitry of the semiconductor die may be in close thermal contact with the crystalline pedestal.

[0011] In a fifth implementation of such a semiconductor device package, the crystalline pedestal may include a single crystal diamond (SCD).

[0012] In a sixth implementation of such a semiconductor device package, the crystalline pedestal may include silicon carbide (SiC) and one or more through-silicon vias (TSVs) disposed within the silicon carbide, wherein each of the one or more through-silicon vias extends in a direction perpendicular to the major plane.

[0013] In accordance with implementations of the subject matter of this disclosure, a method of fabricating a semiconductor device package including a semiconductor die having two opposing surfaces that define a major plane, the method including placing a crystalline pedestal formed of one or more crystals having an anisotropic thermal property above the semiconductor die in thermally conductive, electrically non-conductive contact with a first surface of the semiconductor die, wherein the anisotropic thermal property affects a thermal conductivity of the crystalline pedestal to dissipate heat from the semiconductor die, depositing a first layer of a thermal interface material onto a top surface, opposite from the first surface of the semiconductor die, of the crystalline pedestal, wherein the thermal interface material is thermally conductive and electrically non-conductive, and enclosing at least the semiconductor die, the first layer of the thermal interface material and the pedestal, perpendicular to the major plane, with a packaging lid, wherein the packaging lid is in thermal contact with the first layer of the thermal interface material to promote heat transfer from the crystalline pedestal, through the first layer of the thermal interface material, to the packaging lid.

[0014] In a first implementation of such a method may further include placing a substrate layer adjacent a second surface of the semiconductor die parallel to the major plane and opposite the first surface of the semiconductor die.

[0015] In a second implementation of such a method may further include depositing a second layer of the thermal interface material between (i) the semiconductor die and (ii) the crystalline pedestal to promote transfer of the heat from the semiconductor die to the crystalline pedestal.

[0016] In a third implementation of such a method, placing the crystalline pedestal may include aligning the crystalline pedestal such that the one or more crystals of the crystalline pedestal is oriented to dissipate the heat in a direction extending from the semiconductor die toward the packaging lid.

[0017] According to a first aspect of that third implementation, the placing the crystalline pedestal may include aligning the crystalline pedestal such that the one or more crystals of the crystalline pedestal is oriented in a 111 crystallographic direction.

[0018] In a fourth implementation of such a method may further include thinning the semiconductor die in a direction perpendicular to the major plane to reduce an amount of semiconductor material of the semiconductor die between heat-producing circuitry and the crystalline pedestal.

[0019] In a fifth implementation of such a method, placing the crystalline pedestal above the

semiconductor die may include placing a single crystal diamond above the semiconductor die.

[0020] In a sixth implementation of such a method, placing the crystalline pedestal above the semiconductor die may include placing the crystalline pedestal including silicon carbide, and one or more through-silicon vias disposed within the silicon carbide wherein each of the one or more through-silicon vias extends in a direction perpendicular to the major plane.

[0021] In accordance with implementations of the subject matter of this disclosure, a crystalline pedestal for use within a semiconductor device package including a semiconductor die having two opposing faces that define a major plane, a packaging lid enclosing the semiconductor die, a plurality of contacts on a first exterior surface of the semiconductor device package that is parallel to the major plane, the first exterior surface defining a bottom of the semiconductor device package, and a first layer of a thermal interface material disposed between the crystalline pedestal and the packaging lid, wherein the thermal interface material is thermally conductive and electrically non-conductive, the crystalline pedestal including one or more crystals having an anisotropic thermal property that affects a thermal conductivity of the crystalline pedestal to dissipate heat generated by the semiconductor die when in operation, wherein the crystalline pedestal is disposed above the semiconductor die in thermally-conductive, electrically non-conductive contact with the semiconductor die.

[0022] In a first implementation of such a crystalline pedestal, the one or more crystals of the crystalline pedestal may be oriented to dissipate the heat in a direction extending from the semiconductor die toward the packaging lid.

[0023] According to a first aspect of that first implementation, the one or more crystals of the crystalline pedestal may be oriented in a 111 crystallographic direction.

[0024] In accordance with implementations of the subject matter of this disclosure, a method for conducting heat from a semiconductor die having two opposing surfaces that define a major plane, the method including aligning a crystalline pedestal, formed of one or more crystals having an anisotropic thermal property that affects a thermal conductivity of the crystalline pedestal, above the semiconductor die in thermally conductive, electrically non-conductive contact with a first surface of the semiconductor die, to dissipate the heat from the semiconductor die, depositing a first layer of a thermal interface material onto a top surface, opposite from the first surface of the semiconductor die, of the crystalline pedestal, wherein the thermal interface material is thermally conductive and electrically non-conductive, and enclosing at least the semiconductor die, the first layer of the thermal interface material and the pedestal, perpendicular to the major plane, with a packaging lid, wherein the packaging lid is in thermal contact with the first layer of the thermal interface material to promote heat transfer from the crystalline pedestal, through the first layer of the thermal interface material, to the packaging lid.

[0025] In a first implementation of such a method, aligning the crystalline pedestal may include aligning the crystalline pedestal such that the one or more crystals of the crystalline pedestal is oriented in a 111 crystallographic direction.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0026] Further features of the disclosure, its nature, and various advantages, will be apparent upon consideration of the following detailed description, taken in conjunction with the accompanying drawings, in which like reference characters refer to like parts throughout, and in which:

[0027] FIG. 1 is a vertical cross-sectional view of a semiconductor device package in accordance with implementations of the subject matter of this disclosure;

[0028] FIG. 2 is a vertical cross-sectional view of another semiconductor device package with TSVs in accordance with implementations of the subject matter of this disclosure;

[0029] FIG. 3 is a vertical cross-sectional view of another semiconductor device package with a single crystal diamond in accordance with implementations of the subject matter of this disclosure; and

[0030] FIG. 4 is a flow diagram of a method of fabricating a semiconductor device package in accordance with implementations of the subject matter of this disclosure.

DETAILED DESCRIPTION

[0031] As noted above, a semiconductor die generates heat when in operation, and the package should allow for heat dissipation (e.g., to reduce or avoid hot spots and reduce thermal risk for the die). Moreover, semiconductor dies come in different sizes, including different thicknesses, and it may be advantageous to be able to accommodate different sizes of semiconductor dies in a limited number of semiconductor device package form factors.

[0032] The foregoing objectives may be met, in accordance with implementations of the subject matter of this disclosure, by providing (e.g., in a semiconductor device package) a crystalline pedestal formed of one or more crystals having an anisotropic thermal property which is positioned in a thermally conductive, but electrically nonconductive relationship with the semiconductor die. In addition to serving as a spacer to conform the semiconductor die to a particular semiconductor device package dimension—which may be determined by the dimensions of an apparatus in which the semiconductor device package may be used, or by dimensions of available semiconductor device packages—the crystalline pedestal aids in dissipating, to the ambient environment, heat generated by operation of the semiconductor die. In some implementations, the crystalline pedestal dissipates heat to the ambient environment surrounding the semiconductor device package through thermal interface material (TIM) and a packaging lid of the semiconductor device package.

[0033] The crystalline pedestal positioned above the semiconductor die (considering the substrate of the semiconductor device package to be the bottom) is formed of one or more crystals having an anisotropic thermal property. The anisotropic thermal property of the crystalline pedestal affects thermal conductivity of the crystalline pedestal and the direction of the thermal conductivity (e.g., the direction of heat transfer through the crystalline pedestal). The thermal conductivity of the crystalline pedestal is determinative of the heat dissipation of the crystalline pedestal from the semiconductor die to the ambient environment surrounding the semiconductor device package. In some implementations, the number of crystals, the orientation of each crystal of the crystalline pedestal, and whether the crystalline pedestal includes through-silicon vias (TSVs) may be determined, at least in part, based on the thermal conductivity needed for a respective application of the semiconductor device package. The specific thermal conductivity required for a particular implementation may depend on the expected running temperature of the active semiconductor die and may be determined by simulation or experimentation.

[0034] In addition to having certain anisotropic thermal properties, the crystalline pedestal may be made of a single crystal (e.g., single crystal diamond (SCD)) or more than one crystal (e.g., silicon carbide with TSVs). In some implementations, the TSVs disposed in the silicon carbide may be filled with copper or any other suitable thermally-conductive material to promote heat dissipation from the semiconductor die to the packaging lid of the semiconductor device package.

[0035] In some implementations, each of the crystals of the crystalline pedestal is oriented to promote heat dissipation from the semiconductor die to the ambient environment surrounding the semiconductor device package. The orientation of each crystal of the crystalline pedestal may be referred to by, for example, a Miller index. A Miller index is a set of three integers (e.g., $\langle hkl \rangle$) that may be used to denote an orientation of a respective crystal plane in a lattice, based on intercepts that the crystal plane makes with the crystallographic axes. For example, a Miller index of $\langle hkl \rangle$ includes h —the intercept with the x -axis, k —the intercept with the y -axis, and l —the intercept with the z -axis. A particular thermal conductivity of the crystalline pedestal may be achieved by modifying the orientation of each respective crystal of the crystalline pedestal, causing to change the anisotropic thermal property of the crystalline pedestal. The anisotropic thermal property of the

crystalline pedestal determines, in part, the thermal conductivity of the crystalline pedestal and the direction of the thermal conductivity (e.g., the direction of heat transfer through the crystalline pedestal). In some implementations, each crystal of the crystalline pedestal is oriented in a $\langle 111 \rangle$ crystallographic direction to provide the desired thermal conductivity of the crystalline pedestal. In some implementations, when the crystalline pedestal is made of an SCD, a top surface (e.g., the surface in thermal contact with the TIM layer) is oriented in a $\langle 110 \rangle$ crystallographic direction to provide the desired thermal conductivity of the crystalline pedestal. When the crystalline pedestal is made of an SCD, the thermal conductivity is highest along the $\langle 111 \rangle$ crystallographic direction due to the efficient propagation of phonons along this crystallographic direction. The $\langle 111 \rangle$ crystallographic direction may be defined as the direction extending from one corner of the crystal to the opposite corner, extending through the volumetric center of the crystal.

[0036] The semiconductor device package also includes the packaging lid, which is configured to enclose the semiconductor die, contacts (e.g., an array of bumps, balls, or any other suitable contacts), one or more TIM layers positioned within the semiconductor device package, and the crystalline pedestal.

[0037] The crystalline pedestal should be in thermal contact with the semiconductor die to dissipate heat, but, as noted, should be in an electrically non-conductive contact with the semiconductor die. In one implementation, a TIM, configured to establish a thermally conductive continuum in such implementation, may be disposed between the semiconductor die and the crystalline pedestal. Each layer of TIM positioned within the semiconductor device package should be thermally conductive but electrically non-conductive. Suitable materials that may be used as the TIM include, but are not limited to, silicone adhesives and epoxy adhesives.

[0038] The crystalline pedestal should also be in thermal contact with the packaging lid (if any), configured to enclose the semiconductor die, to promote heat dissipation from the crystalline pedestal to the packaging lid. However, in one implementation, the semiconductor device package further includes a layer of TIM positioned between the crystalline pedestal and the packaging lid.

[0039] In a semiconductor device package implemented with an SCD, there will be TIM positioned between the semiconductor die and the packaging lid, while the SCD will be in direct thermal contact with the semiconductor die. However, in a semiconductor device package implemented with more than one crystal (e.g., SiC crystals), there may also be TIM positioned between the semiconductor die and the crystalline pedestal, to promote heat transfer from the semiconductor die to the crystalline pedestal.

[0040] The subject matter of this disclosure may be better understood by reference to FIGS. 1-4.

[0041] FIG. 1 is a cross-sectional view of a semiconductor device package **100** according to an implementation of the subject matter of this disclosure. Package **100** includes a crystalline pedestal **108** disposed onto a semiconductor die **102** and the semiconductor die **102** separated from a substrate **104** by contacts **106**. Electrical conductivity between semiconductor die **102** and substrate **104** is afforded by contacts **106** or by any other suitable conductive interconnects, which may have cylindrical, spherical, or other suitable shapes. In some implementations, contacts **106** may be formed from a metal such as copper. Contacts **106**, or other suitable connectivity structures, may be used to connect circuit components (not shown) on semiconductor die **102** to exterior package contacts. The package contacts may be pins (not shown), an array of bumps (i.e., a bump array; not shown), or solder balls (i.e., a ball-grid array; not shown). The connections between circuit components on semiconductor die **102** and contacts **106**, and between contacts **106** and package contacts, are not explicitly shown.

[0042] Package **100** further includes a packaging lid **112** that protects the internal elements of package **100**. In some implementations, packaging lid **112** is a lid formed from a thermally-conductive material. In some implementations of this disclosure, the content of package **100** may be completely covered (e.g., by packaging lid **112**). In an enclosed-die package, packaging lid **112** may be integrated with the enclosure or may be a separate lid. The packaging lid **112** may be

formed to leave a void that corresponds to the volume of the package contents, as described below. [0043] If packaging lid **112** is metallic, the metallic lid may be stamped or machined to leave a void of a specific height in package **100**. The specific height may be the total height of contacts **106**, semiconductor die **102**, crystalline pedestal **108**, and thermal interface material (TIM) **114** in which packaging lid **112** would be flush with a top surface of the TIM **114**. In some embodiments, packaging lid **112** (as shown) may have a downward extension **116** that rests on substrate **104**.

[0044] The total height of package **100** is determined by the combined thicknesses of the various layers of package **100** including the thickness of substrate **104**, the thickness of contacts **106**, the thickness of semiconductor die **102**, the thickness of crystalline pedestal **108**, and the thickness of TIM **114**. The thickness of semiconductor die **102** may be determined at least in part by the circuitry to be formed in semiconductor die **102**, which in turn is determined by the intended function of semiconductor die **102**. The thickness of substrate **104** is determined at least in part by the number of layers, which may be a function of electrical requirements, performance requirements, component density, etc.

[0045] The thickness of crystalline pedestal **108** may be determined at least in part by an anisotropic thermal property **110** of the crystals used to form the crystalline pedestal **108**, and the heat to be dissipated, based on the operating temperature of the circuitry on semiconductor die **102**. The anisotropic thermal property **110** of the crystals used to form the crystalline pedestal **108** affects a direction of the transfer of heat through the crystalline pedestal **108**, and thus affects the thermal conductivity of the crystalline pedestal **108**. Therefore, each respective crystal of the crystalline pedestal **108** may be oriented based on the respective anisotropic thermal property **110** of the respective crystal to change the direction of heat transfer and to modify the thermal conductivity of the crystalline pedestal **108**. For example, in an application in which a particular region of semiconductor die **102** runs relatively hotter than other regions of semiconductor die **102**, then the respective crystals closer to the particular region (e.g., in one implementation) are oriented towards the hot spot region so as to provide a directed channel that allows heat to more easily flow away from the particular region. In some implementations, the target size of the overall package **100** also may factor into the thickness of pedestal **108**. In some implementations of the subject matter of this disclosure, the thickness of pedestal **108** may be selected to fill the void, described above, inside package **100** left by a standard packaging lid **112** that has a predetermined height dimension. A semiconductor device package (e.g., semiconductor device package **100**) may be assembled with a standard package size and standard package dimensions, including a standard package height. Such semiconductor device packages (e.g., package **100**) may include semiconductor dies **102** of varying thicknesses, therefore leaving varying void sizes to be filled by one or more internal elements of the package **100**, such as crystalline pedestal **108** and TIM **114**. Therefore, in some implementations, the height of a standard crystalline pedestal **108** may be adapted to fit such a standard packaging lid **112** and provide a thermally conductive continuum, in combination with TIM **114**, by selectively adapting the pedestal thickness, such as by grinding or other techniques for physically altering the pedestal thickness.

[0046] Crystalline pedestal **108** is designed and oriented to exhibit favorable thermally conductive properties to draw heat away from the semiconductor die **102** when semiconductor die **102** is in operation. In some implementations, crystalline pedestal **108** is made of a single crystal (e.g., SCD), multiple crystals (e.g., SiC with TSVs), or any suitable carbon-based crystals.

[0047] As there should be good thermal contact, but not electrical contact, between semiconductor die **102** and crystalline pedestal **108** for the pedestal **108** to serve its function of conducting heat away from semiconductor die **102**, a further layer of TIM (not shown in FIG. 1), which is thermally conductive, but electrically non-conductive, is provided between semiconductor die **102** and crystalline pedestal **108**.

[0048] The crystalline pedestal **108** should also be in thermal contact with packaging lid **112**, configured to enclose the semiconductor die **102**, to promote heat dissipation from the crystalline

pedestal **108** to the packaging lid **112**. The semiconductor device package **100** further includes a layer of TIM **114** positioned between the crystalline pedestal **108** and the packaging lid **112** to ensure such thermal contact between the crystalline pedestal **108** and packaging lid **112** of package **100**, allowing heat to be conducted from pedestal **108** to packaging lid **112** of package **100**, which then conducts or radiates the heat to the ambient environment. As previously noted, a suitable thermally conductive, electrically non-conductive, material to serve as TIM (e.g., TIM **114**) may be silicone adhesive or an epoxy adhesive.

[0049] The inclusion of crystalline pedestal **108** according to an implementation of the subject matter of this disclosure allows the provision of semiconductor device packages (e.g., semiconductor device package **100**) with uniform heights notwithstanding differences in the semiconductor die thickness, by adjusting the thickness of crystalline pedestal **108** to adjust the overall package height.

[0050] In addition to the thermal benefits of including crystalline pedestal **108**, the presence of pedestal **108** allows semiconductor die **102** to be thinner than if there were no pedestal **108**. In some implementations, semiconductor die **102** may be thinned to remove semiconductor material of the semiconductor die **102** between heat-producing circuitry and the crystalline pedestal **108**. The combination of the two thinner layers of crystalline pedestal **108** and semiconductor die **102**, separated by an additional TIM layer (not shown in FIG. **1**), also may reduce mechanical stress on semiconductor die **102** caused by contacts **106**.

[0051] FIG. **2** is a vertical cross-sectional view of another semiconductor device package **200** with multiple crystals **202** (e.g., SiC) and TSVs **204** in accordance with implementations of the subject matter of this disclosure. In some implementations the TSVs **204** disposed in the crystals **202** (e.g., SiC) may be filled with copper or any other suitable thermally-conductive material to promote heat dissipation from the semiconductor die **102** to the packaging lid **112** of the semiconductor device package **200**. In one implementation, the TSVs **204** may be located substantially above hotter running regions of the semiconductor die **102**. In another implementation, the TSVs **204** are uniformly distributed in the crystals **202**, as shown in FIG. **2**.

[0052] In such a semiconductor device package **200** implemented with more than one crystal **202** (e.g., SiC crystal), there may also be TIM **206** positioned between (a) semiconductor die **102** and (b) the crystalline pedestal (e.g., crystals **202** and TSVs **204**), to promote heat transfer from the semiconductor die **102** to the crystalline pedestal (e.g., crystals **202** and TSVs **204**).

[0053] The crystalline pedestal (e.g., crystals **202** and TSVs **204**) should also be in thermal contact with packaging lid **112**, which is configured to enclose the semiconductor die **102**, to promote heat dissipation from the crystalline pedestal (e.g., crystals **202** and TSVs **204**) to the packaging lid **112**. Similar to semiconductor device package **100** of FIG. **1**, semiconductor device package **200** further includes a layer of TIM **114** positioned between the crystalline pedestal (e.g., crystals **202** and TSVs **204**) and the packaging lid **112**. This promotes heat transfer from the crystalline pedestal (e.g., crystals **202** and TSVs **204**) to the packaging lid **112** which then conducts or radiates the heat to the ambient environment.

[0054] In some implementations, the thickness of each TSV **204** may be around 10 μm to 50 μm . However, the thickness of each TSV **204** included in crystals **202** (e.g., SiC crystals) may be based on factors such as the size (e.g., the thickness) of the crystals **202**, a desired thermal performance (e.g., desired thermal conductivity), and the manufacturing process used to fabricate the semiconductor device package **200**.

[0055] FIG. **3** is a vertical cross-sectional view of another semiconductor device package **300** with an SCD **302** of a crystallographic orientation **304** in accordance with implementations of the subject matter of this disclosure.

[0056] Similarly to semiconductor device package **100** in FIG. **1**, semiconductor device package **300** implemented with SCD **302** includes TIM **114** between SCD **302** and the packaging lid **112**, while SCD **302** will be in direct thermal contact with the semiconductor die **102**.

[0057] In accordance with implementations of the present disclosure, SCD **302** is oriented to promote heat dissipation from semiconductor die **102** to the ambient environment surrounding the semiconductor device package **300**. A particular thermal conductivity of the SCD **302** may be achieved by modifying the orientation **304** of SCD **302**. For example, SCD **302** shown in FIG. **3** is oriented in a $\langle 111 \rangle$ crystallographic direction to establish the anisotropic thermal property **110** and provide the desired thermal conductivity of the SCD **302**. In one implementation, the SCD **302** can be oriented to draw heat from one or more particular regions of semiconductor die **102** that runs hotter relative to other regions of semiconductor die **102**.

[0058] Method **400** in accordance with implementations of the subject matter of this disclosure for fabricating a semiconductor device package is diagrammed in FIG. **4**. At **402**, a crystalline pedestal formed of one or more crystals having an anisotropic thermal property is placed above the semiconductor die to be in thermally conductive, electrically non-conductive contact with a first surface of the semiconductor die. At **404**, a first layer of a thermal interface material is deposited onto a top surface (e.g., the major face opposite from the first surface of the semiconductor die) of the crystalline pedestal, wherein the thermal interface material is thermally conductive and electrically non-conductive. At **406**, at least the semiconductor die, the first layer of the thermal interface material, and the pedestal are enclosed with the packaging lid, perpendicular to the major plane, wherein the packaging lid is in thermal contact with the thermal interface layer to promote heat transfer from the crystalline pedestal, through the thermal interface material, to the packaging lid. Method **400** then ends.

[0059] Thus it is seen that a semiconductor device package is provided with a crystalline pedestal formed of at least one crystal having an anisotropic thermal property, which promotes the dissipation of heat generated by the semiconductor die when in operation.

[0060] As used herein and in the claims which follow, the construction “one of A and B” shall mean “A or B.”

[0061] It is noted that the foregoing is only illustrative of the principles of the invention, and that the invention can be practiced by other than the described embodiments, which are presented for purposes of illustration and not of limitation, and the present invention is limited only by the claims which follow.

Claims

1. A semiconductor device package comprising: a semiconductor die having two opposing faces that define a major plane, wherein the semiconductor die generates heat when in operation; a packaging lid enclosing the semiconductor die; a plurality of contacts on a first exterior surface of the semiconductor device package that is parallel to the major plane, the first exterior surface defining a bottom of the semiconductor device package; a crystalline pedestal formed of one or more crystals having an anisotropic thermal property affecting a thermal conductivity of the crystalline pedestal to dissipate the heat generated by the semiconductor die when in operation, the crystalline pedestal disposed above the semiconductor die in thermally-conductive, electrically non-conductive contact with the semiconductor die; and a first layer of a thermal interface material disposed between the pedestal and the packaging lid, wherein the thermal interface material is thermally conductive and electrically non-conductive.
2. The semiconductor device package of claim 1, further comprising a substrate layer below the semiconductor die and forming the first exterior surface.
3. The semiconductor device package of claim 1, further comprising: a second layer of the thermal interface material disposed between (i) the semiconductor die and (ii) the crystalline pedestal, wherein the second layer of the thermal interface material is configured to promote transfer of the heat from the semiconductor die to the crystalline pedestal.
4. The semiconductor device package of claim 1, wherein the one or more crystals of the crystalline

pedestal is oriented to dissipate the heat in a direction extending from the semiconductor die toward the packaging lid.

5. The semiconductor device package of claim 4, wherein the one or more crystals of the crystalline pedestal is oriented in a 111 crystallographic direction.

6. The semiconductor device package of claim 1, wherein heat-producing circuitry of the semiconductor die is in close thermal contact with the crystalline pedestal.

7. The semiconductor device package of claim 1, wherein the crystalline pedestal comprises a single crystal diamond.

8. The semiconductor device package of claim 1, wherein the crystalline pedestal comprises: silicon carbide; and one or more through-silicon vias disposed within the silicon carbide, wherein each of the one or more through-silicon vias extends in a direction perpendicular to the major plane.

9. A method of fabricating a semiconductor device package comprising a semiconductor die having two opposing surfaces that define a major plane, the method comprising: placing a crystalline pedestal formed of one or more crystals having an anisotropic thermal property above the semiconductor die in thermally conductive, electrically non-conductive contact with a first surface of the semiconductor die, wherein the anisotropic thermal property affects a thermal conductivity of the crystalline pedestal to dissipate heat from the semiconductor die; depositing a first layer of a thermal interface material onto a top surface, opposite from the first surface of the semiconductor die, of the crystalline pedestal, wherein the thermal interface material is thermally conductive and electrically non-conductive; and enclosing at least the semiconductor die, the first layer of the thermal interface material and the pedestal, perpendicular to the major plane, with a packaging lid, wherein the packaging lid is in thermal contact with the first layer of the thermal interface material to promote heat transfer from the crystalline pedestal, through the first layer of the thermal interface material, to the packaging lid.

10. The method of fabricating a semiconductor device package according to claim 9, the method further comprising placing a substrate layer adjacent a second surface of the semiconductor die parallel to the major plane and opposite the first surface of the semiconductor die.

11. The method of fabricating a semiconductor device package according to claim 9, the method further comprising depositing a second layer of the thermal interface material between (i) the semiconductor die and (ii) the crystalline pedestal to promote transfer of the heat from the semiconductor die to the crystalline pedestal.

12. The method of fabricating a semiconductor device package according to claim 9, wherein the placing the crystalline pedestal comprises aligning the crystalline pedestal such that the one or more crystals of the crystalline pedestal is oriented to dissipate the heat in a direction extending from the semiconductor die toward the packaging lid.

13. The method of fabricating a semiconductor device package according to claim 12, wherein the placing the crystalline pedestal comprises aligning the crystalline pedestal such that the one or more crystals of the crystalline pedestal is oriented in a 111 crystallographic direction.

14. The method of fabricating a semiconductor device package according to claim 9, the method further comprising thinning the semiconductor die in a direction perpendicular to the major plane to reduce an amount of semiconductor material of the semiconductor die between heat-producing circuitry and the crystalline pedestal.

15. The method of fabricating a semiconductor device package according to claim 9, wherein placing the crystalline pedestal above the semiconductor die comprises placing a single crystal diamond above the semiconductor die.

16. The method of fabricating a semiconductor device package according to claim 9, wherein placing the crystalline pedestal above the semiconductor die comprises placing the crystalline pedestal comprising: silicon carbide; and one or more through-silicon vias disposed within the silicon carbide wherein each of the one or more through-silicon vias extends in a direction

perpendicular to the major plane.

17. A crystalline pedestal for use within a semiconductor device package including a semiconductor die having two opposing faces that define a major plane, a packaging lid enclosing the semiconductor die, a plurality of contacts on a first exterior surface of the semiconductor device package that is parallel to the major plane, the first exterior surface defining a bottom of the semiconductor device package, and a first layer of a thermal interface material disposed between the crystalline pedestal and the packaging lid, wherein the thermal interface material is thermally conductive and electrically non-conductive, the crystalline pedestal comprising: one or more crystals having an anisotropic thermal property that affects a thermal conductivity of the crystalline pedestal to dissipate heat generated by the semiconductor die when in operation; wherein: the crystalline pedestal is disposed above the semiconductor die in thermally-conductive, electrically non-conductive contact with the semiconductor die.

18. The crystalline pedestal of claim 17, wherein the one or more crystals of the crystalline pedestal are oriented to dissipate the heat in a direction extending from the semiconductor die toward the packaging lid.

19. The crystalline pedestal of claim 18, wherein the one or more crystals of the crystalline pedestal is oriented in a 111 crystallographic direction.

20. A method for conducting heat from a semiconductor die having two opposing surfaces that define a major plane, the method comprising: aligning a crystalline pedestal, formed of one or more crystals having an anisotropic thermal property that affects a thermal conductivity of the crystalline pedestal, above the semiconductor die in thermally conductive, electrically non-conductive contact with a first surface of the semiconductor die, to dissipate the heat from the semiconductor die; depositing a first layer of a thermal interface material onto a top surface, opposite from the first surface of the semiconductor die, of the crystalline pedestal, wherein the thermal interface material is thermally conductive and electrically non-conductive; and enclosing at least the semiconductor die, the first layer of the thermal interface material and the pedestal, perpendicular to the major plane, with a packaging lid, wherein the packaging lid is in thermal contact with the first layer of the thermal interface material to promote heat transfer from the crystalline pedestal, through the first layer of the thermal interface material, to the packaging lid.

21. The method of for conducting heat from the semiconductor die according to claim 20, wherein aligning the crystalline pedestal comprises aligning the crystalline pedestal such that the one or more crystals of the crystalline pedestal is oriented in a 111 crystallographic direction.
